Syntax of Arrow Functions

An arrow function is created by using the `=>` syntax, replacing the function keyword in function expressions.

Basic Syntax:

```
const functionName = (parameters) => {
  // Function body
};
```

If the function has only one parameter, parentheses can be omitted, and if it has only one expression, you can omit the curly braces `{}` and the `return` keyword — it will return the result of that expression by default.

Examples of Declaring and Using Arrow Functions

No Parameters

When there are no parameters, you need to include empty parentheses `()`.

```
const greet = () => console.log("Hello, World!");
greet(); // Output: Hello, World!
```

Single Parameter

With a single parameter, you can omit the parentheses.

```
const greetUser = name => console.log(`Hello, ${name}!`);

greetUser('Alice'); // Output: Hello, Alice!
```

Multiple Parameters

For multiple parameters, parentheses are required.

```
const add = (a, b) => a + b;

console.log(add(3, 4)); // Output: 7
```

Multi-line Function Body

For more complex functions with multiple lines, you use curly braces {} and explicitly return the value if needed.

```
const multiplyLogAndReturn = (a, b) => {
  const result = a * b;
  console.log(`The result is: ${result}`);
  return result;
};

let x = multiplyLogAndReturn(3, 5); // Output: The result is: 15
console.log(`x is: ${x}`);          // Output: x is: 15
```

~ End ~